COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 07

Combinational Circuit

By- Chandan Gupta Sir



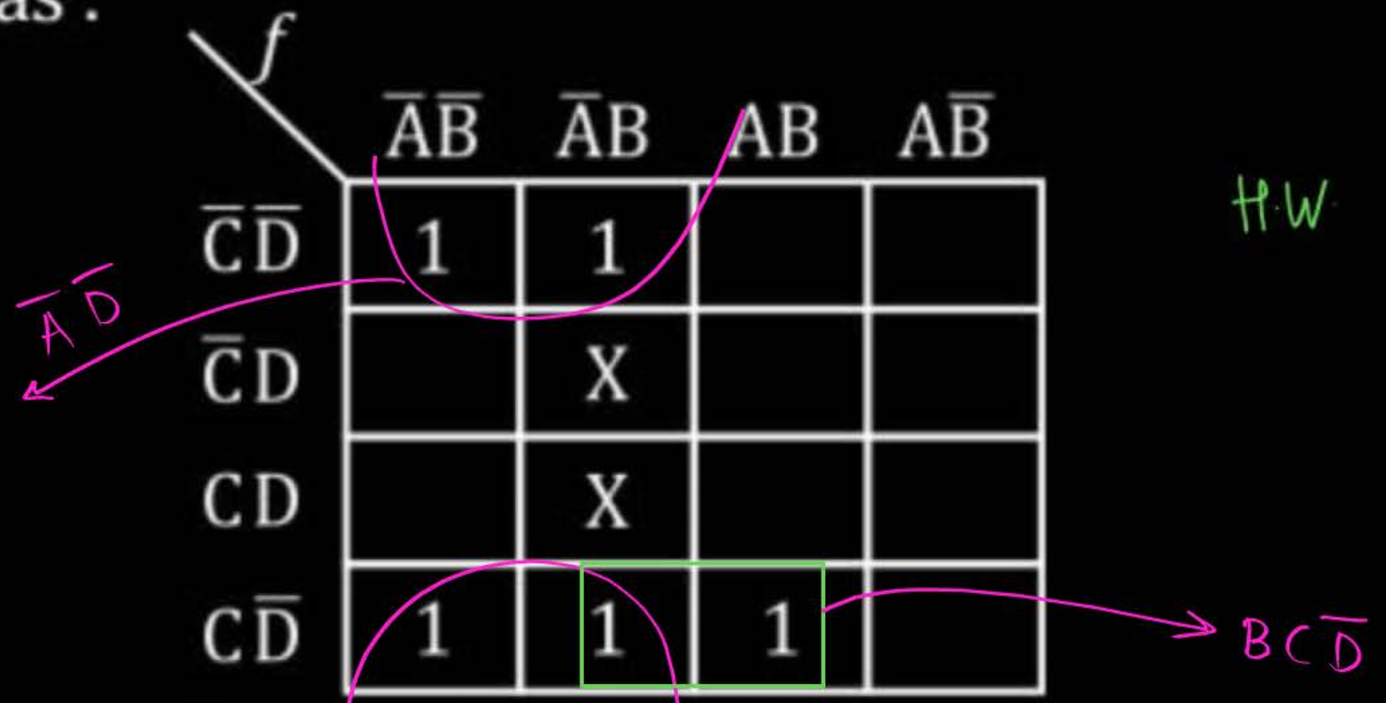
K-Map.



K-Map. Cont.

Question

A K-map is given as :



The minimized expression of above K-map is :

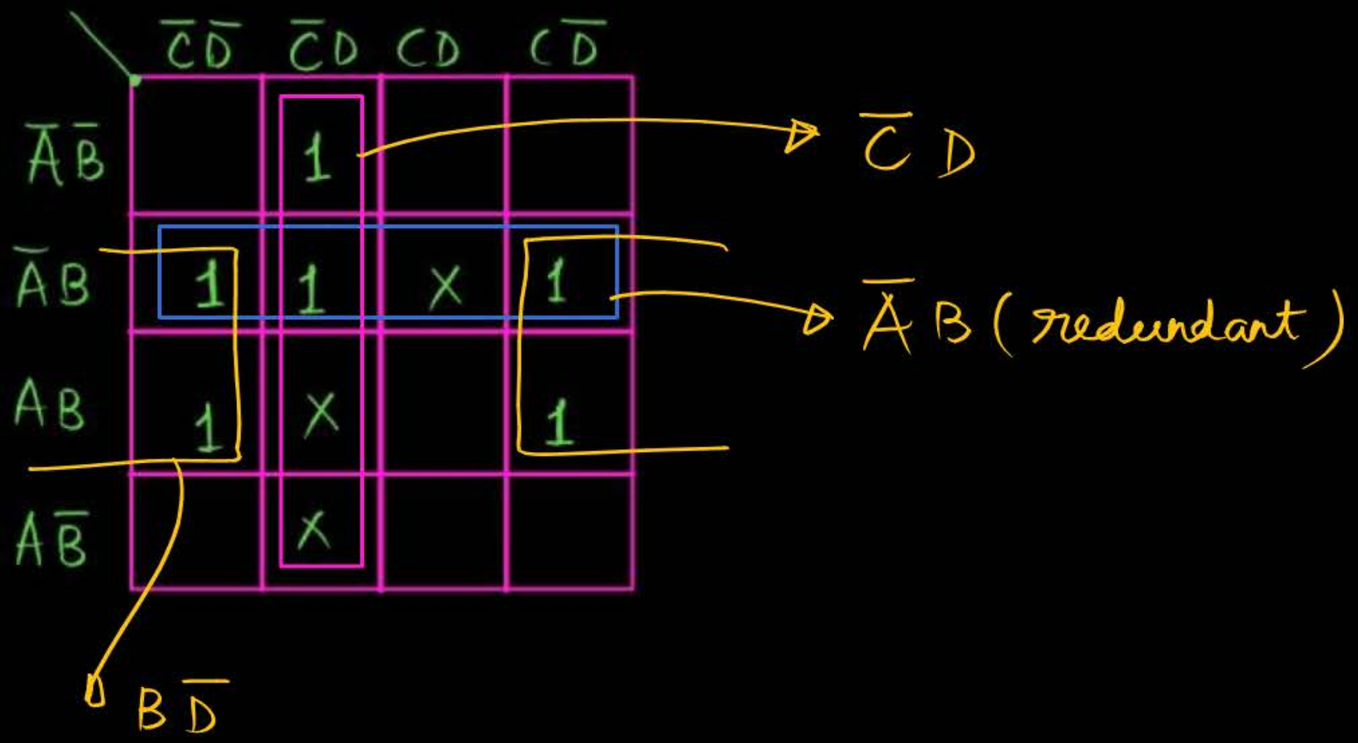
(a) $\bar{A}B + \bar{A}\,\bar{D} + BC\bar{D}$

(b) $\bar{A}\,\bar{D} + BC\bar{D}$

(c) $\bar{A}\bar{B}\bar{D} + \bar{A}B + BC\bar{D}$

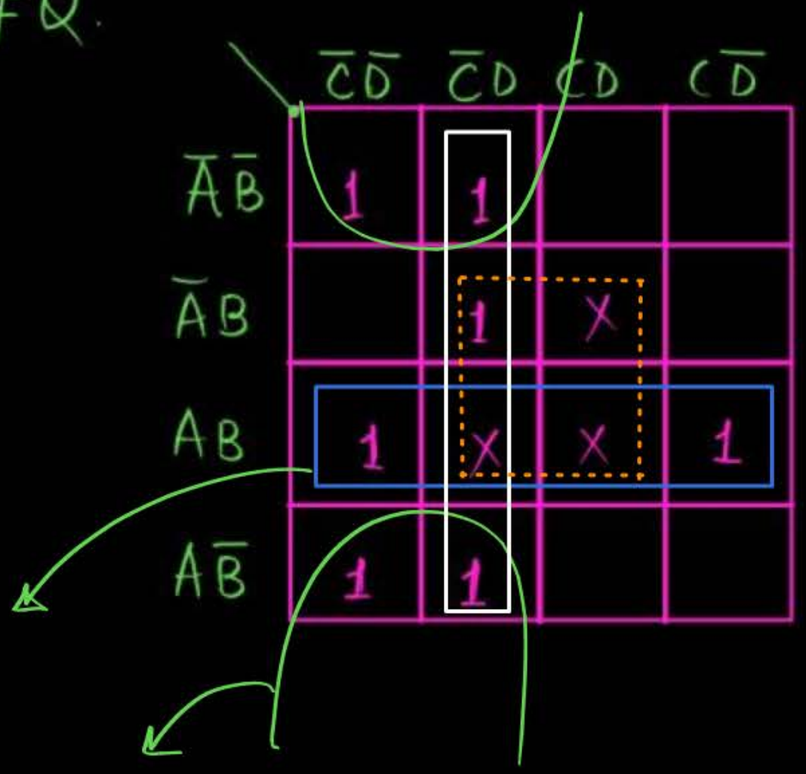
(d) None of these

#Q.1.



$= B\bar{D} + \bar{C}D$

#Q.

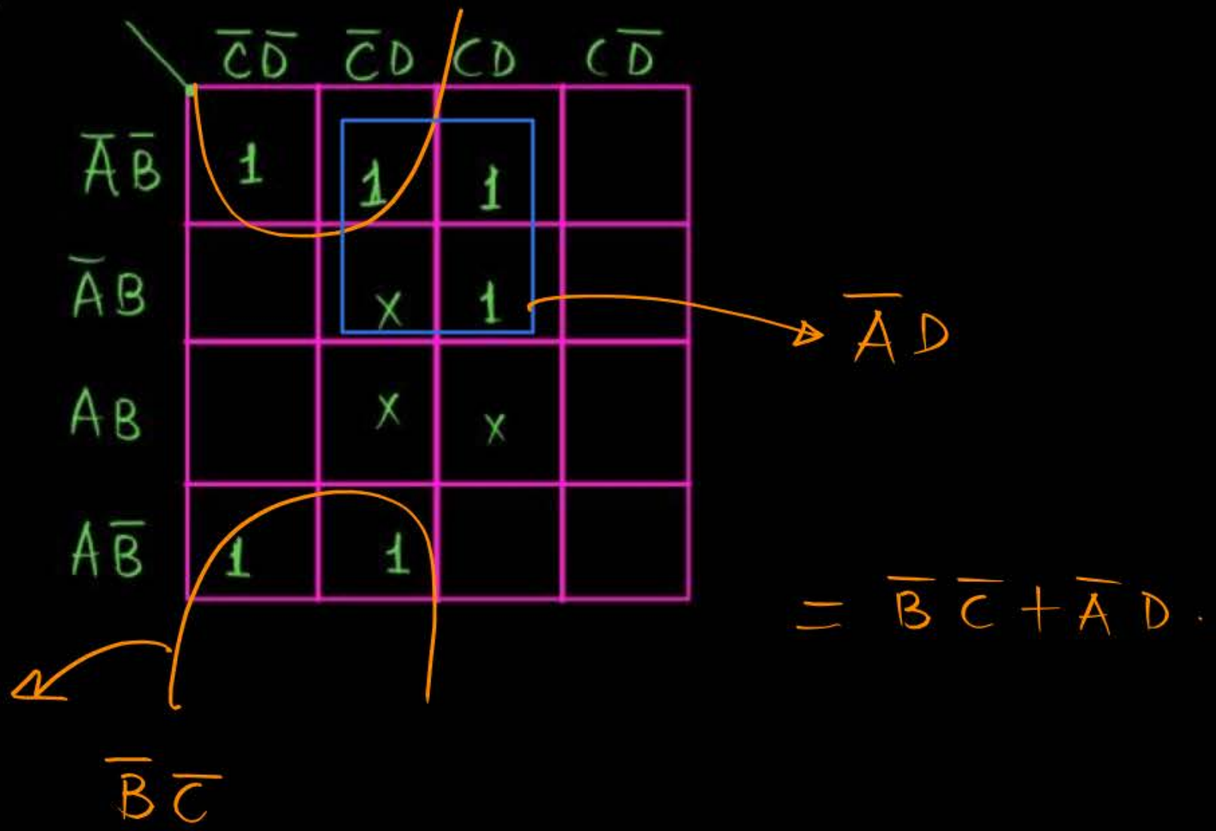


$\bar{B}\bar{C} + AB + \bar{C}D$

OR

$\bar{B}\bar{C} + AB + BD.$

#Q.



$= \bar{B}\bar{C} + \bar{A}D.$

Question

$f(A, B, C) = A\bar{C} + A\bar{B}C + BC$

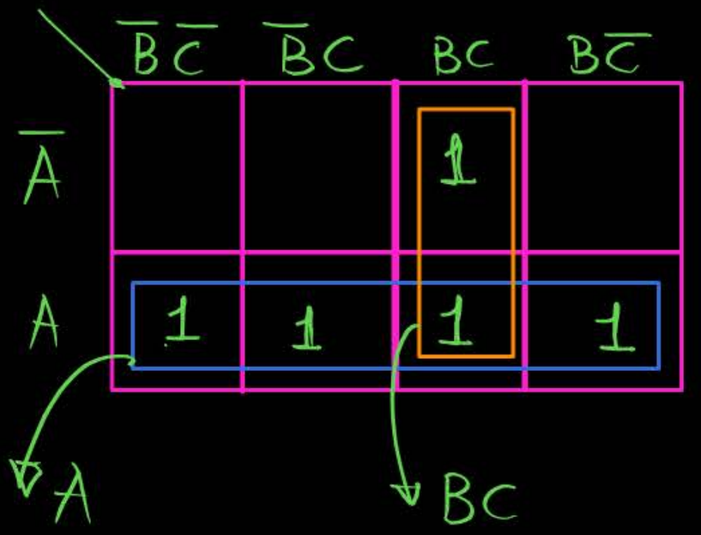
Then its minimized expression will be

(a) $(A + B)(A + C) = A + BC$

(b) $A \odot C + BC$

(c) $\bar{A} + BC$

(d) None of these



Question

$$f(A, B, C, D) = \bar{A}\,\bar{D} + BC\bar{D} + AC$$

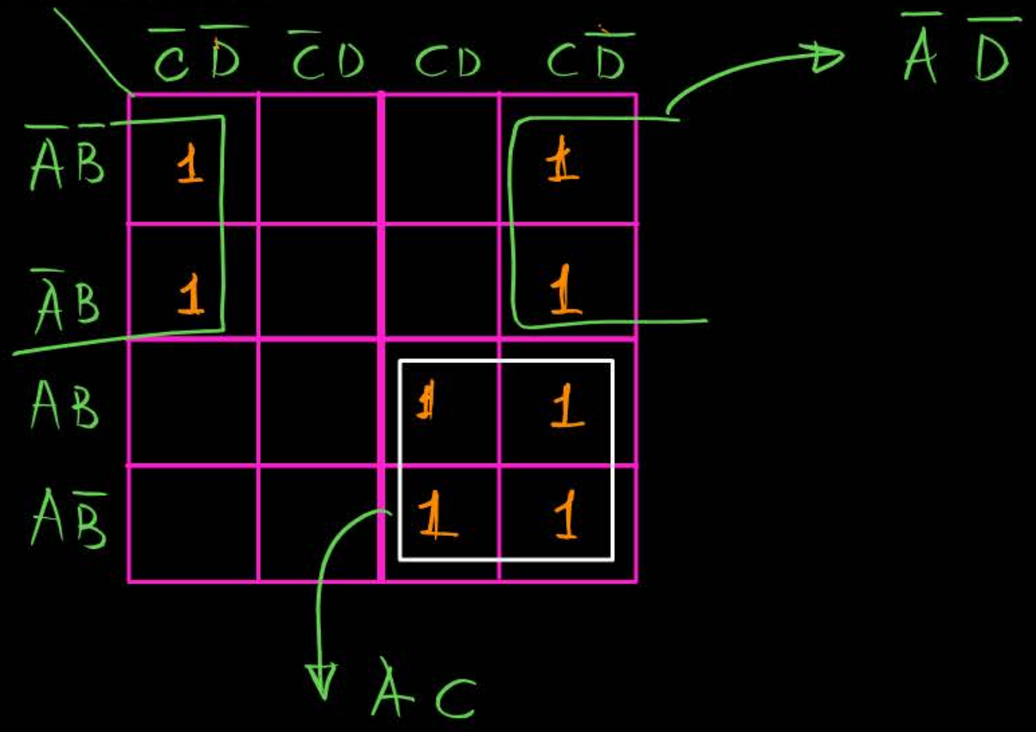
Then its minimized expression will be

(a) $\bar{A}\,\bar{D} + AC$

(b) $\bar{A}\,\bar{D} + BC$

(c) $BC + AC$

(d) $\bar{A}\,\bar{D} + C\bar{D}$



Question

$$f(A,B,C) = \overline{\bar{A}C + B\bar{C}} + AB\bar{C} \quad = \overline{f_1(A,B,C)} + AB\bar{C}$$

Then its minimized expression will be

(a) $(A+\bar{B})(A+\bar{C})$

(b) $A\bar{C} + \bar{B}C$

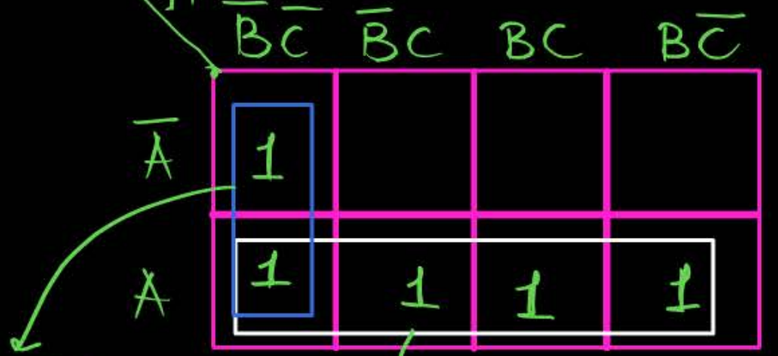
(c) $\bar{A}\bar{B} + A\bar{C}$

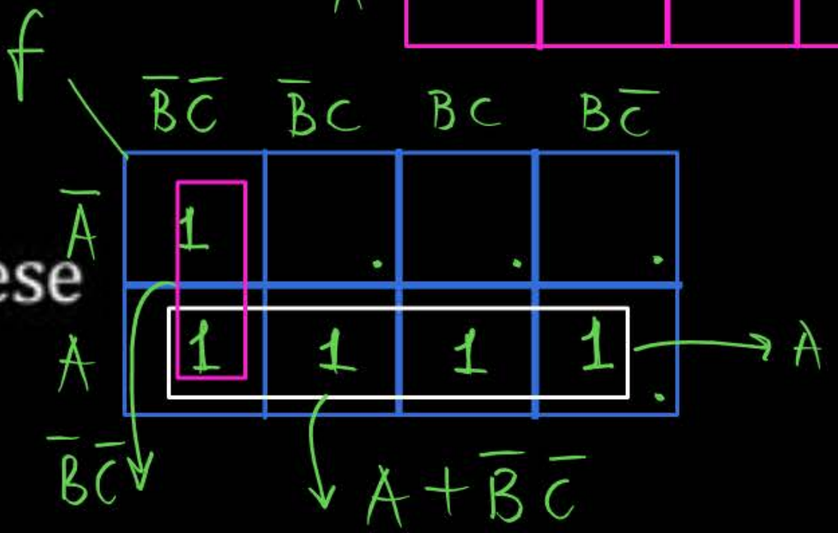
(d) None of these



$f_1(A,B,C) = \bar{A}C + B\bar{C}$

$\bar{f}_1 + AB\bar{C}$





$= A + \bar{B}\bar{C}$

Question

$$f(A,B,C,D) = \overline{\bar{C}D + A\bar{B}D + AC} + \bar{A}\bar{B}C$$

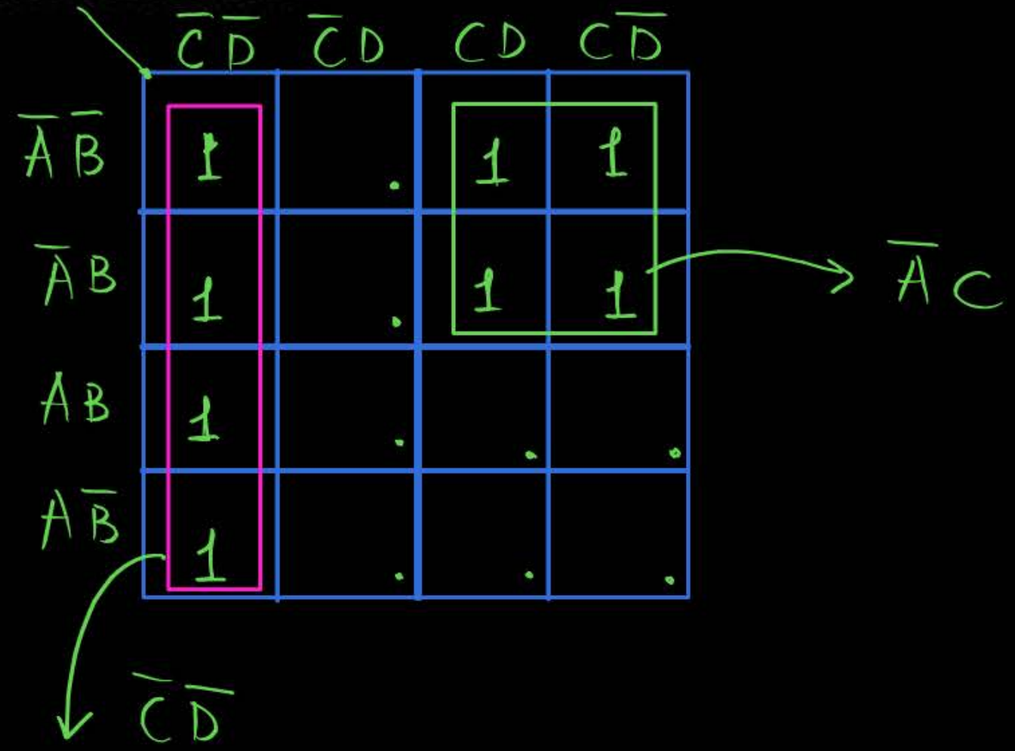
Then its minimized solution will be

(a) $\bar{A}\bar{C}D + \bar{B}\bar{D}$

(b) $\bar{C}\,\bar{D} + \bar{A}C$

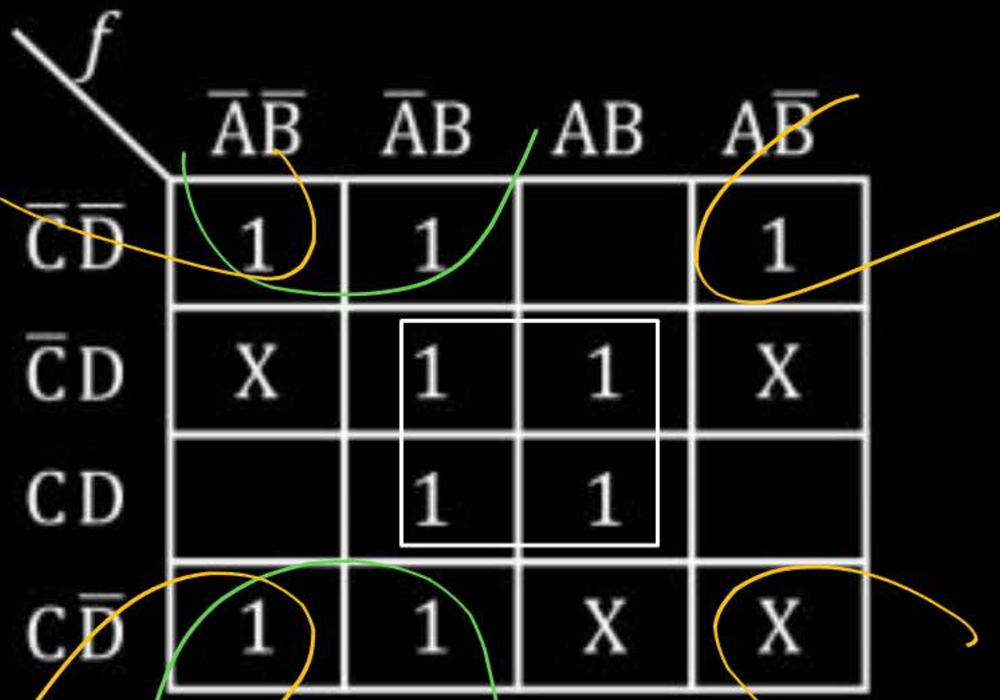
(c) $\bar{A}\bar{B} + \bar{A}\,\bar{D} + \bar{C}D$

(d) None of these



Question

A K-map is given as :



H.W.

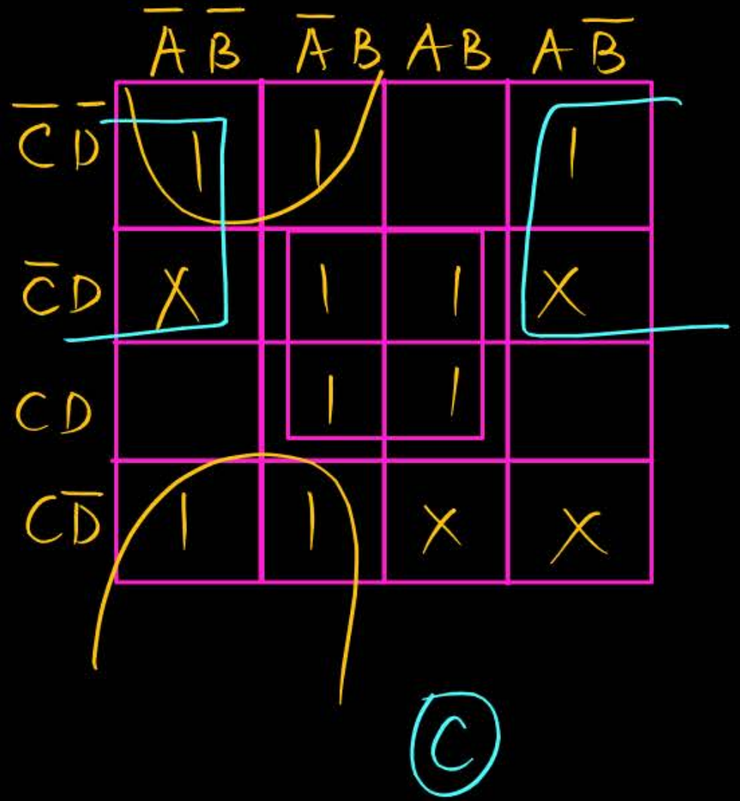
Then which of the following is/are minimized form of above K-map ?

(a) $\bar{A}B + C\bar{D} + BD + \bar{B}\bar{D}$

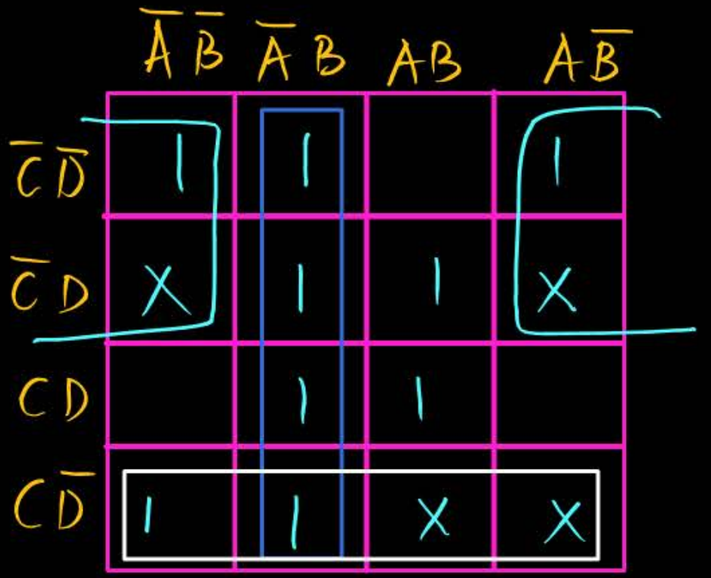
(b) $BD + \bar{A}\bar{D} + \bar{B}\bar{D}$

(c) $\bar{A}\bar{D} + BD + \bar{B}\bar{C}$

(d) $\bar{B}\bar{C} + \bar{A}B + C\bar{D}$


$\bar{A}\bar{B}$ $\bar{A}B$ AB $A\bar{B}$
$\bar{C}\bar{D}$ 1 1 1
$\bar{C}D$ X 1 1 X
CD 1 1
$C\bar{D}$ 1 1 X X

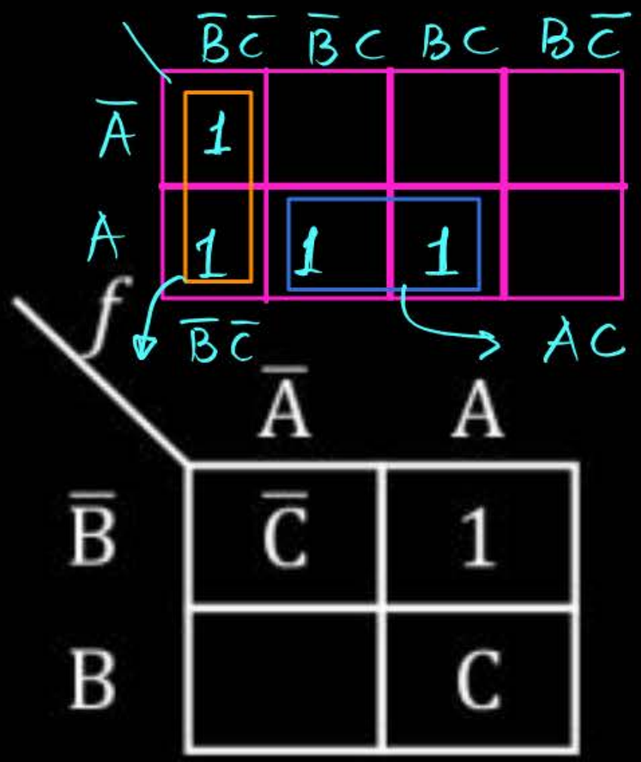
(c)


$\bar{A}\bar{B}$ $\bar{A}B$ AB $A\bar{B}$
$\bar{C}\bar{D}$ 1 1 1
$\bar{C}D$ X 1 1 X
CD 1 1
$C\bar{D}$ 1 1 X X

(d) X

Question

A K-map is given as :



$f(A,B,C)$

$= \bar{A}\bar{B}\cdot\bar{C} + A\bar{B}\cdot 1 + \bar{A}B\cdot 0 + ABC\cdot$

The solution of the K-map will be :

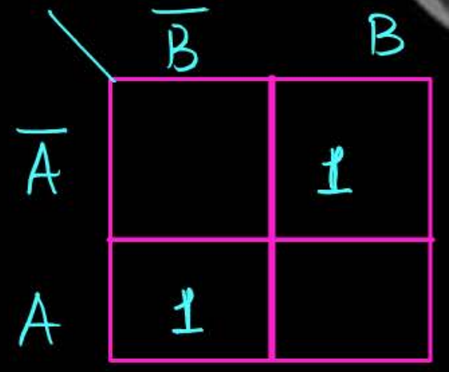
(a) $\bar{B}\bar{C} + AC$

(b) $\bar{A}\bar{B} + AC$

(c) $\bar{A}B + B\bar{C}$

(d) None of these

$f_1(A,B) = \Sigma(1,2) = \bar{A}B + A\bar{B}$



$= \bar{A}\bar{B}\cdot 0 + \bar{A}B\cdot 1 + A\bar{B}\cdot 1 + AB\cdot 0$

$= \bar{A}B + A\bar{B}$

$f_1(0,0) = 0$ $f_1(1,1) = 0$

$f_1(0,1) = 1$

$f_2(1,0) = 1$

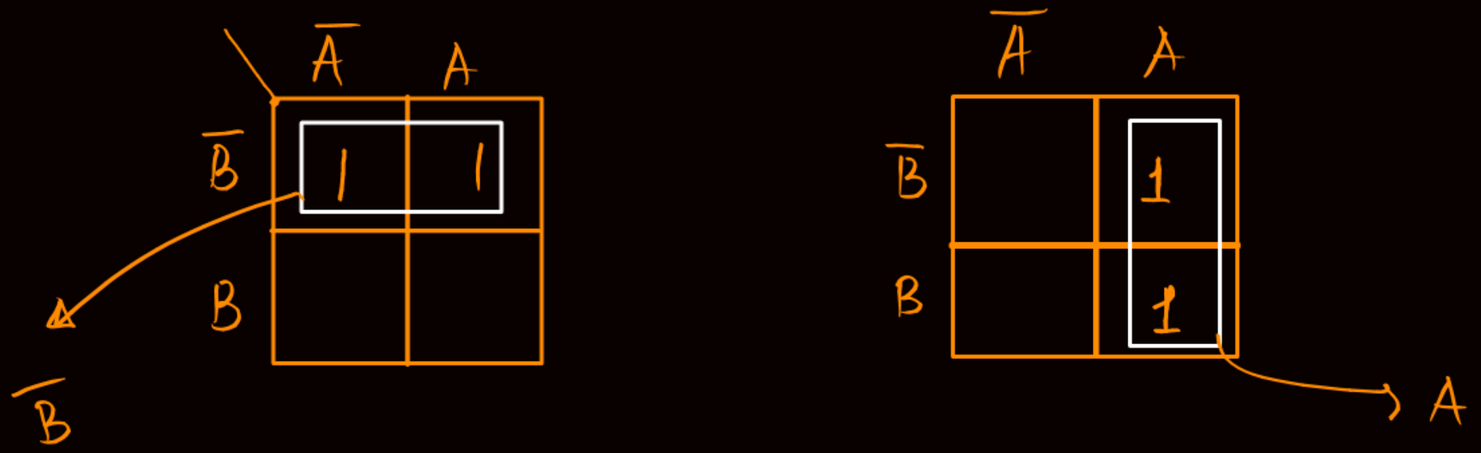
$$f(A,B,C) = \bar{A}\bar{B}\bar{C} + A\bar{B} + ABC$$

$$f(0,0,C) = \bar{C} + 0 + 0 = \bar{C}$$

$$f(1,0,C) = 0 + 1 + 0 = 1$$

$$f(0,1,C) = 0$$

$$f(1,1,C) = 0 + 0 + C = C$$

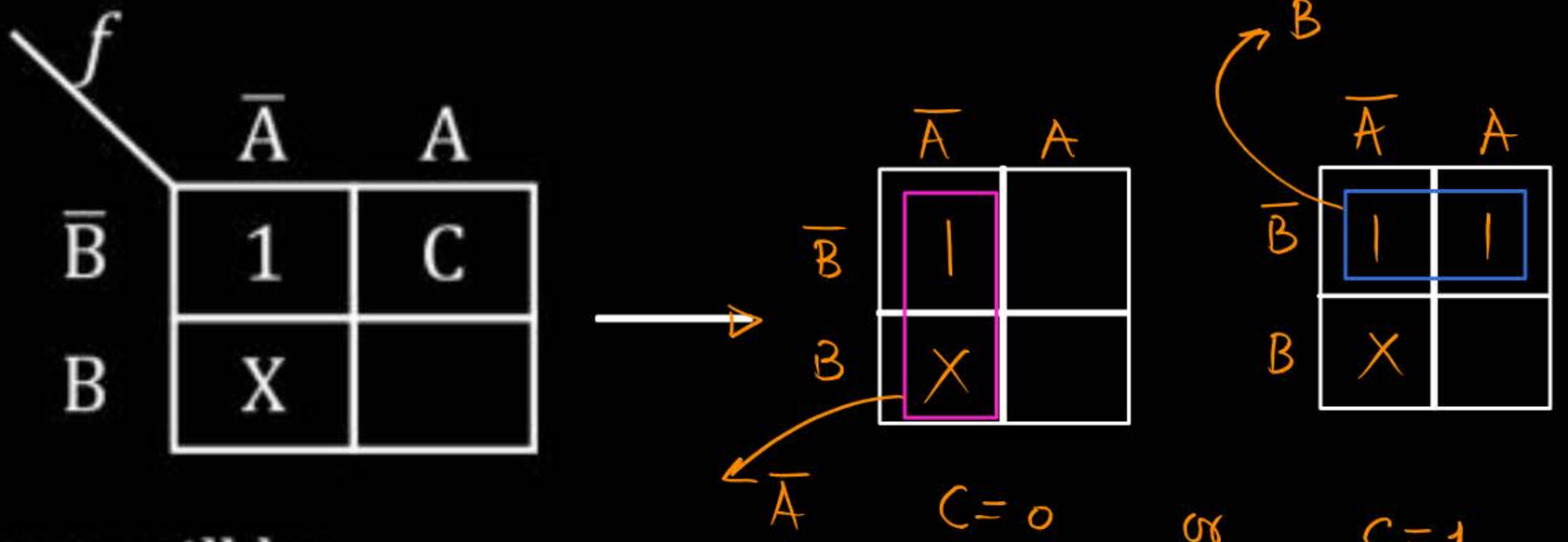


$C = 0$ or $C = 1$

$$= \overline{C} \cdot \overline{B} + C \cdot A$$

Question

A K-map is given as :



The solution of the K-map will be :

(a) $\bar{A}\bar{C} + \bar{B}\bar{C}$

(b) $AC + \bar{B}\bar{C}$

(c) $\bar{A}\bar{C} + \bar{B}C$

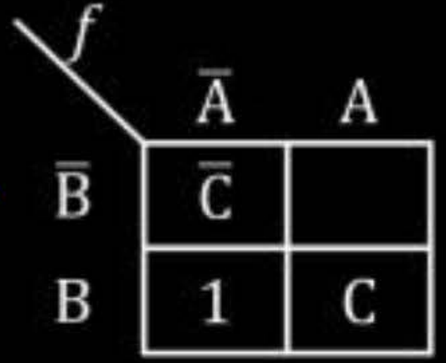
(d) None of these

Question

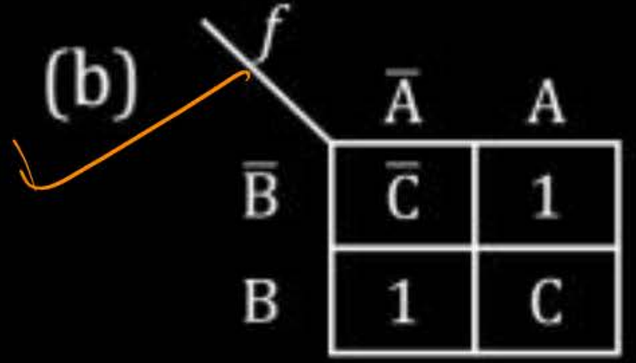
$$f(A, B, C) = A\bar{B} + BC + \bar{A}\bar{C}$$

Then the K-map implementing above logical function will be :

(a)

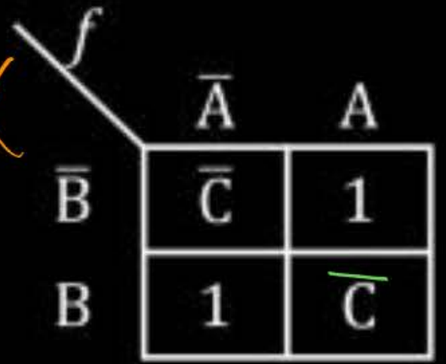


(b)

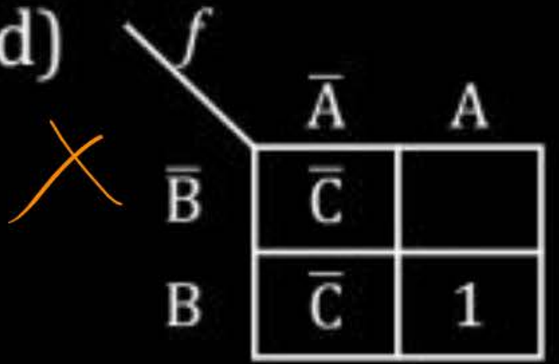


$f(1,0,C)$
$= 1$
$f(1,1,C)$
$= 0 + C + 0$
$= C$

(c)



(d)



Question

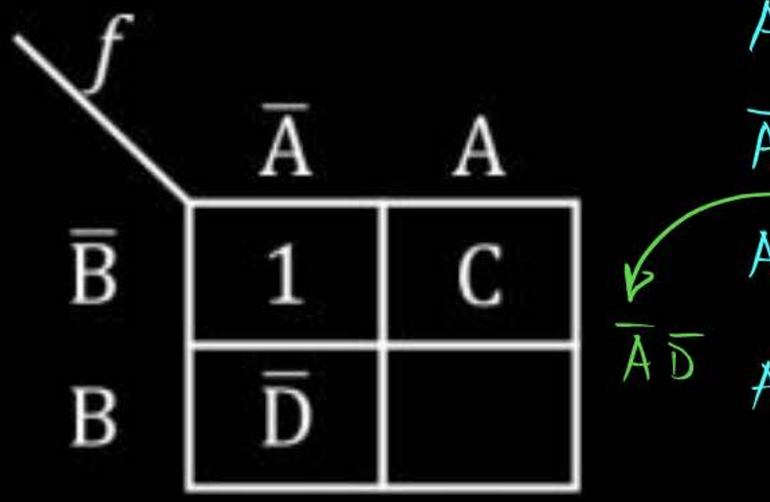
K-map of $f(A, B, C, D)$ is given as :

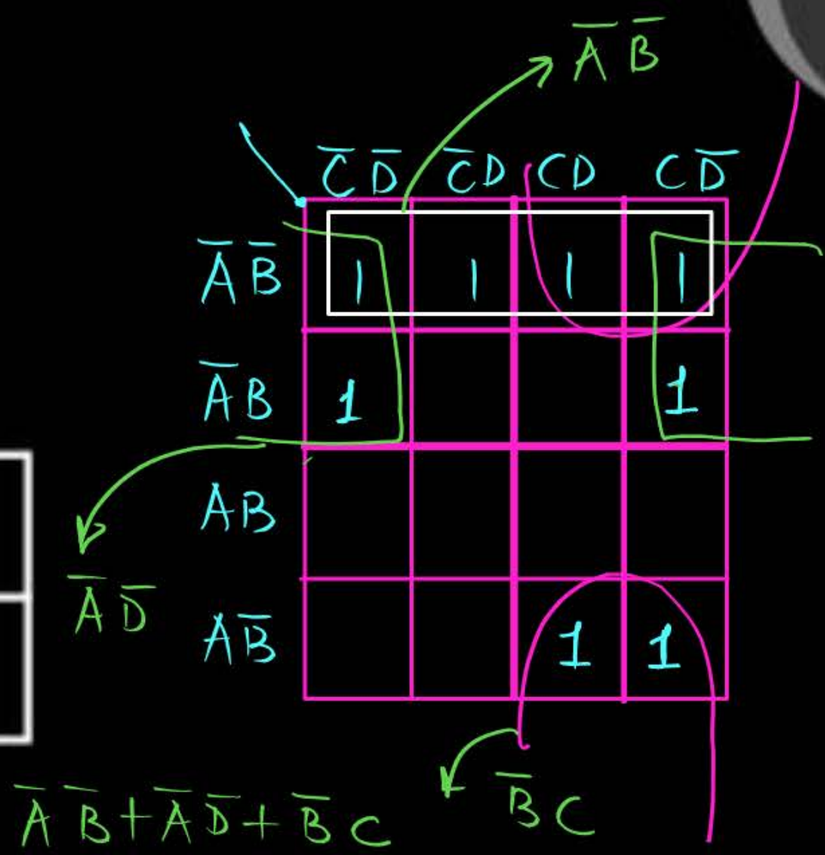
$f(A,B,C,D) = \bar{A}\bar{B}\cdot 1 + A\bar{B}\cdot C + \bar{A}B\bar{D}$

$f(0,0,C,D) = 1$

$f(1,0,C,D) = C$

Then $f(A, B, C, D)$ will be :



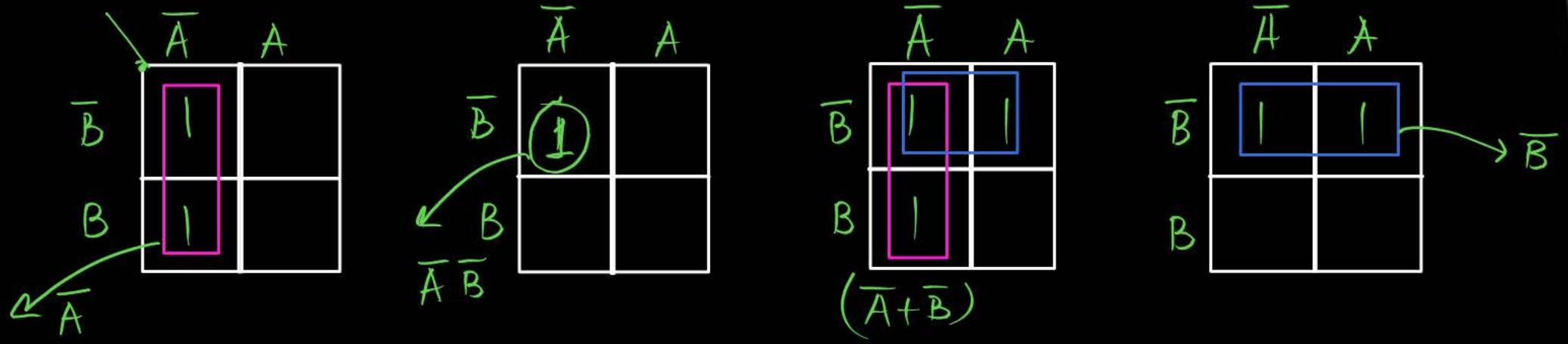


(a) $\bar{A}\bar{C}\bar{D} + \bar{A}C\bar{D} + \bar{B}C\bar{D}$

(b) $\bar{A}\bar{B} + \bar{A}\bar{D} + \bar{B}C$

(c) $\bar{A}\bar{C} + \bar{B}\bar{D} + BC$

(d) $\bar{A}\bar{B} + \bar{B}\bar{D} + BC$



C=0 & D=0 or C=0 & D=1 or C=1 & D=1 or C=1 & D=0

$\bar{A}C\bar{D} + \bar{B}C\bar{D}$

$$= \bar{C}\bar{D}\cdot\bar{A} + \bar{C}D\bar{A}\bar{B} + C\bar{D}\cdot(\bar{A}+\bar{B}) + C\cdot D\cdot(\bar{B})$$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	1
$\bar{A}B$	1			1
AB				
$A\bar{B}$			1	1

Question

If the function W, X, Y, Z are as :

$$W = R + \bar{P}Q + \bar{R}S \qquad X = PQ\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S}$$

$$Y = RS + \overline{PR + P\bar{Q} + \bar{P}\bar{Q}}$$

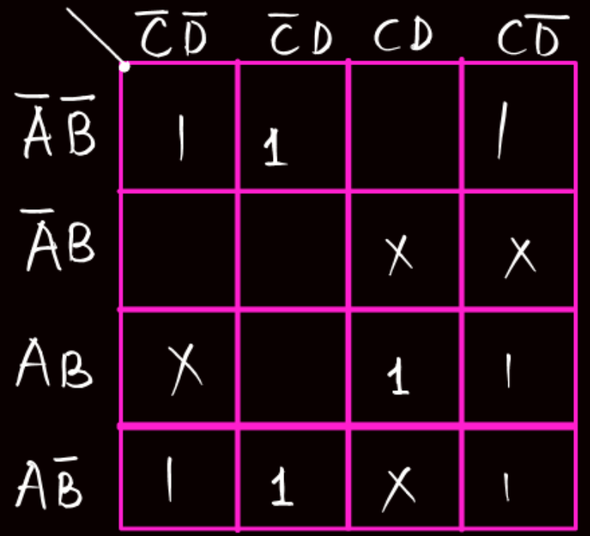
$$Z = R + S + \overline{PQ + \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{S}}$$

Then which of the following is/are correct ?

(a) $W = \bar{X}$ (b) $W = \bar{Z}$

(c) $\bar{W} = \bar{Z}$ (d) $\bar{X} = Z$

Q.



	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1		1
$\bar{A}B$			X	X
AB	X		1	1
$A\bar{B}$	1	1	X	1

#Q. $f(A,B,C,D) = \overline{\bar{A}\bar{D} + \bar{B}\bar{C} + AB\bar{C}} + B\bar{D}$

Minimize this boolean function using K-Map.

#

$f(A,B,C) = AB + BC + CA = \Sigma(3,5,6,7)$

$f(1,1,0) = 1 \quad , \quad f(0,0,1) = 0$

$f(\bar{A}, B, C) = \bar{A}B + BC + \bar{A}C$

$f(AB, BC, CA) = ABC + ABC + ABC = ABC$

$f[f(A,B,B), B, C] = \Sigma(2,3,6,7) = \pi(0,1,4,5)$

$f(B,B,C) = B$

$f(A,B,B) = AB + B + AB$

$= B$



2 Minute Summary

K-Map

Thank you

GW

Soldiers !

